

Empty is OFF. A dot is ON. Pressing a line changes BOTH lamps at its ends: OFF becomes ON, and ON becomes OFF.

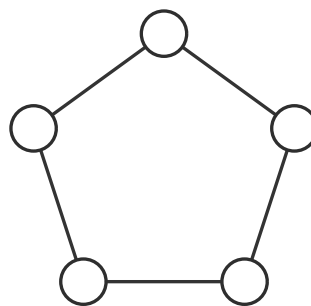
Problem 1: Do-nothing moves

Start all OFF. Choose at least one line. Press each chosen line once. Can you finish with every lamp OFF again?

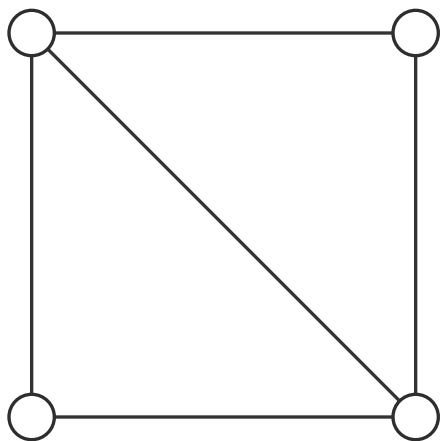
A.



B.



C. Find EVERY nonempty set of lines that changes nothing.



What do all the answers have in common?

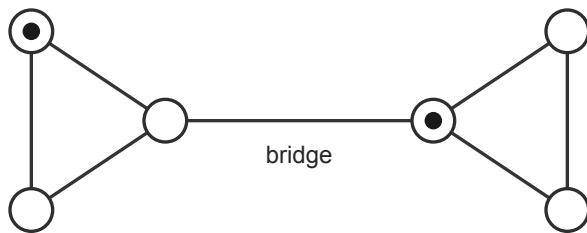
Problem 2: Bridge detective

For this problem, press each line at most once.

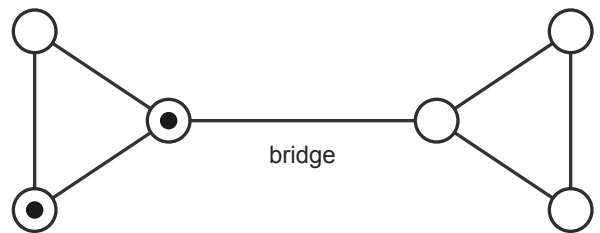
Before solving each puzzle, must you press the bridge, or must you leave it alone? How can you know without solving the whole puzzle? Then solve to check your prediction.

A.

Start

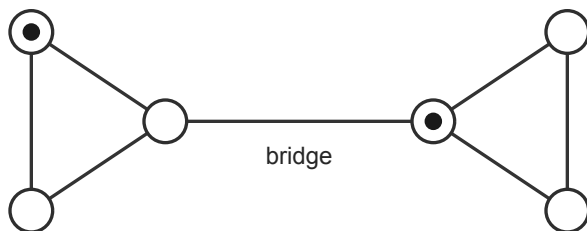


Target

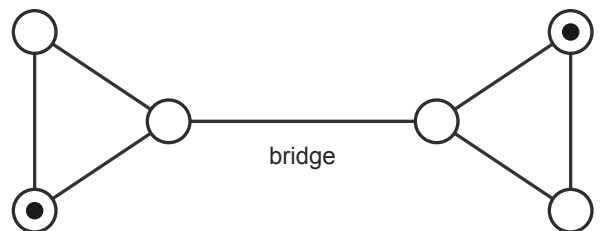


B.

Start



Target

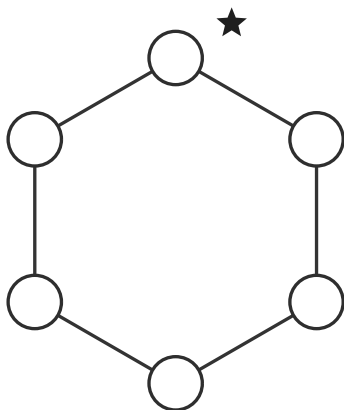


Problem 3: One solo switch

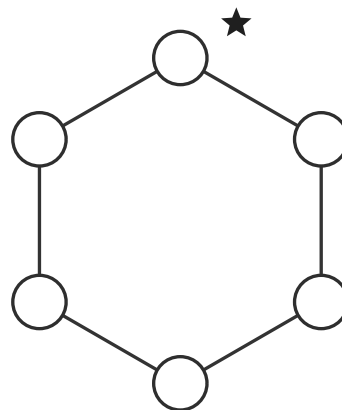
NEW RULE - this problem only: The lamp with a star has a special button. The button changes ONLY its own lamp. Lines still change the two lamps at their ends.

Start all OFF each time.

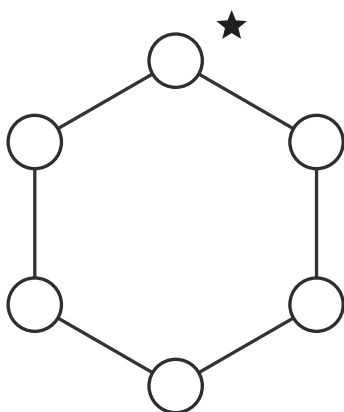
A. Make exactly the special lamp ON.



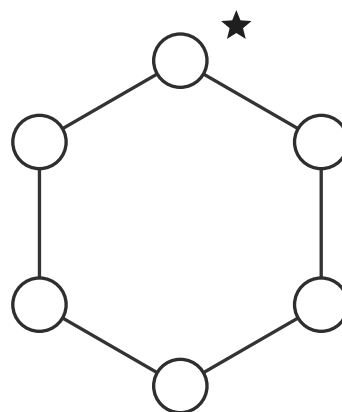
B. Make exactly a neighboring lamp ON.



C. Make exactly a lamp far away ON.



D. Make any target you want.

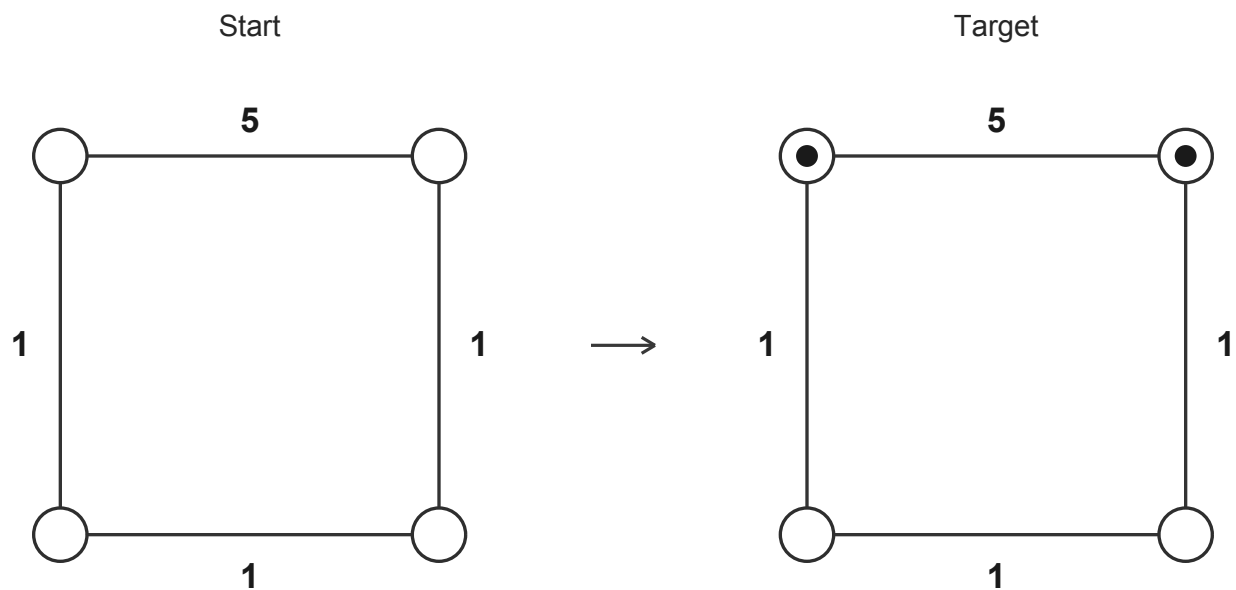


With the special button, can you make EVERY target picture?

Problem 4: Cheapest, not fewest

Start all OFF. Make the target picture.

A press costs the number written on its line.



a. What solution uses the FEWEST presses?

b. What solution costs the LEAST?

c. Why are the answers different?
